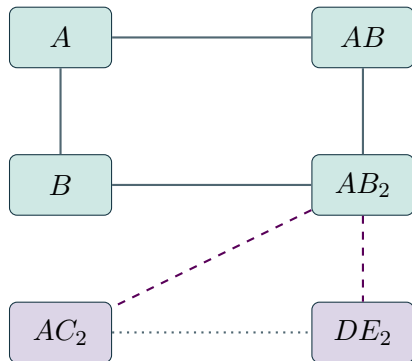


Retrieved reference graph

Query composition: AB_2



Each node is a whole crystal

Solid: chemical relation Dashed: analogue relation

Dotted: structural similarity

Light blue: Competition references

Light purple: Cross-chemistry analogues

$$x_r = [h_r^{\text{Crys-JEPA}}; c_r]$$

Frozen crystal embedding h_r

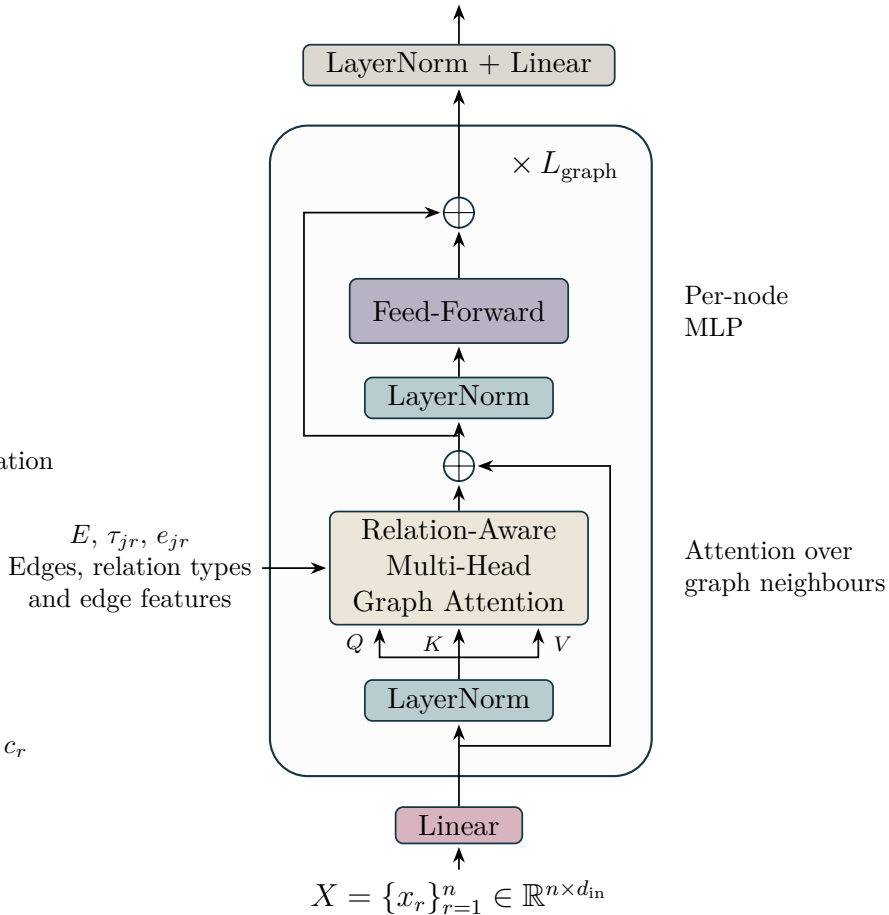
Element identities and composition fractions c_r

Analogue references provide context;
they are not decomposition products.

(a) Reference inputs

One contextual token per reference

$$R = \{R_r\}_{r=1}^n \in \mathbb{R}^{n \times d_{\text{ref}}}$$



No positional encoding; no global pooling

(b) Proposed reference graph encoder